

Corrigé

Partie 1

Exercice 1 — $U_1 = 5, q = 2 \ U_2 = 10, U_3 = 20, U_4 = 40, U_5 = 80$

Exercice 2 — $U_1 = 4, q = 3 \ U_2 = 12, U_3 = 36, U_4 = 108$

Exercice 3 — $U_1 = 80, q = 0,5 \ U_2 = 40, U_3 = 20, U_4 = 10, U_5 = 5$

Exercice 4 — $U_1 = -3, q = -2 \ U_2 = 6, U_3 = -12, U_4 = 24, U_5 = -48.$

Partie 2

Exercice 5 — $U_1 = 3, q = 2 \ U_9 = U_1 \times q^8 = 3 \times 256 = \mathbf{768}$

Exercice 6 — $U_1 = 100, q = 0,1 \ U_7 = U_1 \times q^6 = 100 \times 0,000001 = \mathbf{0,0001}$

Exercice 7 — $U_4 = 24, q = 2 \ U_4 = U_1 \times q^3 \Rightarrow 24 = U_1 \times 8 \Rightarrow U_1 = 3 \ U_{11} = U_1 \times q^{10} = 3 \times 1024 = \mathbf{3072}$

Exercice 8 — $U_1 = 2, q = -3 \ U_6 = U_1 \times q^5 = 2 \times (-243) = -486 \ U_7 = U_1 \times q^6 = 2 \times 729 = 1458$
 U_6

Partie 3

Exercice 9 — $U_1 = 2, q = 3, \text{ somme de 6 termes } S_6 = U_1 \times (1 - q^6)/(1 - q) = 2 \times (1 - 729)/(1 - 3) = 2 \times (-728)/(-2) = \mathbf{728}$

Exercice 10 — $U_1 = 1, q = 2, \text{ somme de 10 termes } S_{10} = 1 \times (1 - 2^{10})/(1 - 2) = (1 - 1024)/(-1) = \mathbf{1023}$

Exercice 11 — $U_1 = 5, q = 0,5, \text{ somme de 7 termes } S_7 = 5 \times (1 - 0,5^7)/(1 - 0,5) = 5 \times (1 - 0,0078125)/0,5 = 5 \times 1,984375 = \mathbf{9,921875}$

Exercice 12 (synthèse) — $U_1 = 3, q = -2$

1. $U_1 = 3, U_2 = -6, U_3 = 12, U_4 = -24, U_5 = 48$

2. $U_8 = U_1 \times q^7 = 3 \times (-128) = \mathbf{-384}$

3. $S_8 = U_1 \times (1 - q^8)/(1 - q) = 3 \times (1 - 256)/(1 - (-2)) = 3 \times (-255)/3 = \mathbf{-255}$